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Business Requirements Document

Defining Needs and Objectives for Project Success

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1. Introduction

This Business Requirements Document (BRD) outlines the key objectives, needs, and expectations for the proposed project. It serves as a guide for stakeholders, ensuring alignment and clarity throughout the project lifecycle.

2. Purpose

The purpose of this document is to define the business requirements needed for the project to succeed. It will provide a foundation for design, development, and implementation activities.

3. Background

NeuByte developed this fitness-tracking platform to illustrate how everyday user experiences can be transformed through thoughtful design, strong data architecture, and actionable insights. The project demonstrates NeuByte's ability to translate business needs into a cohesive technical solution, integrating cloud services, structured data models, and intuitive reporting. It serves as a representative example of NeuByte's approach to building reliable, scalable systems that empower users and support informed decision-making.

4. Objectives

1. Project Objectives

- Demonstrate NeuByte's end-to-end execution of the full Software Development Life Cycle through the delivery of a functional fitness-tracking application.
- Highlight the documentation, analysis, design, development, testing, and support practices that reflect NeuByte's standards for clarity, structure, and technical excellence.
- Provide a user-centered solution that supports personal fitness goals through intuitive workflows and accessible data insights.

2. Objectives

- Enable users to create, update, and manage personalized exercise programs.
- Allow users to log exercise metrics associated with their selected programs, including sets, reps, duration, and performance details.
- Provide users with the ability to track body weight and other personal metrics over time.
- Deliver periodic progress reporting (weekly, monthly, annually) to support long-term goal tracking and data-driven decision-making.

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5. Scope

In Scope

The Fitness Tracking Application will include the following capabilities:

- User Authentication & Account Management
  - Secure user registration, login, logout, and basic profile management.
- Exercise Program Management
  - Creation, editing, and deletion of personalized exercise programs, including program metadata and associated exercises.
- Activity & Metric Logging
  - Entry, modification, and deletion of workout metrics such as sets, reps, duration, weight used, and performance notes.
- Personal Health Tracking
  - Recording and tracking of body weight and other basic personal metrics over time.
- Data Storage & Retrieval
  - Persistent storage of user data using a cloud-based database, with structured retrieval for reporting and analytics.
- Reporting & Insights
  - Generation of weekly, monthly, and annual progress summaries based on logged data and exercise program activity.
- Notifications & System Alerts
  - Basic system-generated notifications related to data entry, program updates, or reporting availability.
- User Interface (UI)
  - A responsive, user-friendly interface that supports core workflows and provides clear navigation across features.

Out of Scope

The following items are not included in the initial release:

- Integration with third-party fitness devices or wearables
- Social features such as friend lists, leaderboards, or shared programs
- Advanced analytics, predictive modeling, or AI-driven recommendations
- Mobile-native applications (iOS/Android)
- Multi-tenant administration or enterprise-level RBAC
- Offline data synchronization
- Nutrition tracking or meal planning features.

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6. Requirements

Business Requirements

Login Flow

BR-LOGIN-FLOW-01
The system shall enforce the following login sequence:
CAPTCHA → Credentials → MFA → Dashboard.

BR-LOGIN-FLOW-02
The system shall prevent users from bypassing any step in the login sequence.

BR-LOGIN-FLOW-03
The system shall redirect users to the Dashboard upon successful completion of MFA.

BR-LOGIN-FLOW-04
The system shall log all authentication events (CAPTCHA pass/fail, login success/failure, MFA success/failure).

Login Captcha

BR-LOGIN-CAPTCHA-01
The system shall require users to complete a CAPTCHA challenge before allowing access to the login form.

BR-LOGIN-CAPTCHA-02
The system shall prevent access to the username/password fields until the CAPTCHA is successfully validated.

BR-LOGIN-CAPTCHA-03
The system shall display an error message if the CAPTCHA validation fails.

BR-LOGIN-CAPTCHA-04
The system shall limit repeated failed CAPTCHA attempts to mitigate automated attacks.

Login

BR-Login-01: Display Login Page
The system shall display a login page that includes fields for entering a valid email address and password.

BR-Login-02: Validate Required Fields
The system shall validate that both the email and password fields are populated before allowing the user to submit the login form.

BR-Login-03: Validate Email Format
The system shall validate that the email address entered conforms to standard email formatting rules.

BR-Login-04: Authenticate User Credentials
The system shall authenticate the user by verifying the provided email and password against stored credentials in the database.

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BR-Login-05: Password Hash Verification
The system shall compare the submitted password to the stored password using secure hashing.

BR-Login-06: Display Authentication Errors
The system shall display an error message when authentication fails due to invalid credentials.

BR-Login-07: Successful Login Redirect
The system shall redirect authenticated users to the application's home/dashboard page.

BR-Login-08: Session Creation
The system shall create

a secure session token upon successful login.

BR-Login-09: Session Storage
The system shall store the session token in a secure, HTTP-only cookie.

BR-Login-10: Session Timeout
The system shall automatically expire user sessions after a defined period of inactivity.

BR-Login-11: Logout Functionality
The system shall allow users to log out, which terminates the active session and removes the session cookie.

BR-Login-12: Access Control
The system shall restrict access to authenticated pages unless a valid session token is present.

BR-Login-13: Error Messaging for Locked or Disabled Accounts
The system shall display a specific error message if a user attempts to log in with an account that is locked, disabled, or inactive.

BR-Login-14: Prevent Multiple Failed Attempts
The system shall track failed login attempts and temporarily lock the account after a defined threshold is exceeded.

BR-Login-15: Secure Transmission
The system shall transmit all login credentials and session tokens over HTTPS.

BR-Login-16: Provide Create Account Navigation
The system shall provide a link on the login page that navigates users to the account creation page.

BR-Login-17: Provide Reset Password Navigation
The system shall provide a link on the login page that navigates users to the password reset page.

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BR-Login-18: Display Email Format Error
The system shall display a specific error message when the email address does not meet standard formatting rules.

BR-Login-19: Display Required Field Errors
The system shall display an error message when required fields (email or password) are left blank.

BR-Login-20: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon on the login page.

BR-Login-21: Mask Password Input
The system shall mask password characters as they are entered.

BR-Login-22: Provide Accessible Labels
The system shall provide accessible labels for all input fields to support screen readers.

BR-Login-23: Keyboard Navigation Support
The system shall allow users to navigate the login form using keyboard input only.

BR-LOGIN-MFA-01
The system shall require MFA for all users after successful primary authentication.

BR-LOGIN-MFA-02
The system shall support MFA via an Authenticator App (TOTP).

BR-LOGIN-MFA-03
The system shall support MFA via Email One-Time Code (OTP).

BR-LOGIN-MFA-04
The system shall allow users to complete MFA using either TOTP or Email OTP based on their configured preference.

BR-LOGIN-MFA-05
The system shall display an error message when an invalid MFA code is entered.

BR-LOGIN-MFA-06
The system shall expire MFA codes after a defined time period.

BR-LOGIN-MFA-07
The system shall allow users to request a new MFA code when using Email OTP.

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Create Account

BR-Create-01: Display Create Account Page
The system shall display a Create Account page containing fields for Email, Password, and Confirm Password.

BR-Create-02: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon on the Create Account page.

BR-Create-03: Required Fields Validation
The system shall require the Email, Password, and Confirm Password fields to be populated before allowing account creation.

BR-Create-04: Validate Email Format
The system shall validate that the email address entered conforms to standard email formatting rules.

BR-Create-05: Password Complexity Validation
The system shall validate that the password meets the minimum complexity requirements:

- At least 8 characters
- At least 1 number
- At least 1 special character

BR-Create-06: Confirm Password Match
The system shall validate that the Confirm Password field matches the Password field.

BR-Create-07: Display Field-Level Validation Errors
The system shall display specific error messages when validation fails (e.g., invalid email, weak password, mismatched passwords).

BR-Create-08: Check for Existing Account
The system shall prevent account creation if the email address is already associated with an existing user account.

BR-Create-09: Password Hashing
The system shall securely hash the password before storing it in the database.

BR-Create-10: Create User Account Record
The system shall create a new user account record upon successful validation of all fields.

BR-Create-11: Display Account Creation Errors
The system shall display an error message if account creation fails due to system or database errors.

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BR-Create-12: Successful Account Creation Redirect
The system shall redirect the user to the Login page after successful account creation.

BR-Create-13: Mask Password Fields
The system shall mask characters entered in the Password and Confirm Password fields.

BR-Create-14: Secure Transmission
The system shall transmit all account creation data over HTTPS.

BR-Create-15: Provide Accessible Labels
The system shall provide accessible labels for all input fields to support screen readers.

BR-Create-16: Keyboard Navigation Support
The system shall allow users to navigate the Create Account form using keyboard input only.

Reset Password

BR-Reset-01: Display Reset Password Page
The system shall display a Reset Password page containing fields for Email, Password, and Confirm Password.

BR-Reset-02: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon on the Reset Password page

BR-Reset-03: Required Fields Validation
The system shall require the Email, Password, and Confirm Password fields to be populated before allowing a password reset.

BR-Reset-04: Validate Email Format
The system shall validate that the email address entered conforms to standard email formatting rules.

BR-Reset-05: Password Complexity Validation
The system shall validate that the new password meets the minimum complexity requirements:

- At least 8 characters
- At least 1 number
- At least 1 special character

BR-Reset-06: Confirm Password Match
The system shall validate that the Confirm Password field matches the Password field.

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BR-Reset-07: Display Field-Level Validation Errors
The system shall display specific error messages when validation fails (e.g., invalid email, weak password, mismatched passwords).

BR-Reset-08: Verify Account Exists
The system shall verify that the provided email address is associated with an existing user account.

BR-Reset-09: Prevent Reuse of Previous Ten Passwords
The system shall prevent users from resetting their password to one of their previous ten used passwords.

BR-Reset-10: Password Hashing
The system shall securely hash the new password before storing it in the database.

BR-Reset-11: Update User Password
The system shall update the user's stored password upon successful validation of all fields.

BR-Reset-12: Display Reset Errors
The system shall display an error message if the password reset fails due to system or database errors.

BR-Reset-13: Successful Reset Redirect
The system shall redirect the user to the Login page after a successful password reset.

BR-Reset-14: Mask Password Fields
The system shall mask characters entered in the Password and Confirm Password fields.

BR-Reset-15: Secure Transmission
The system shall transmit all password reset data over HTTPS.

BR-Reset-16: Provide Accessible Labels
The system shall provide accessible labels for all input fields to support screen readers.

BR-Reset-17: Keyboard Navigation Support
The system shall allow users to navigate the Reset Password form using keyboard input only.

Dashboard

BR-DASH-01: Display Dashboard
The system shall display a dashboard page immediately after successful user authentication.

BR-DASH-02: Display User Identity
The system shall display the logged-in user's email address on the dashboard.

BR-DASH-03: Display Last Login Timestamp
The system shall display the user's last login date and time.

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BR-DASH-04: Display Selected Exercise Program
The system shall display the name of the user's currently selected exercise program.

BR-DASH-05: Display Program Progress
The system shall display the user's current week and/or day within the selected exercise program, when applicable.

BR-DASH-06: Display Current Weight
The system shall display the user's most recently recorded weight.

BR-DASH-07: Display Weight Trend
The system shall calculate and display the user's weight trend (increase, decrease, or no change) based on the most recent entries.

BR-DASH-08: Display Today's Workout Summary
The system shall display a summary of the workout scheduled for the current day, including exercise count and estimated duration.

BR-DASH-09: Start Workout Action
The system shall provide a "Start Workout" button that navigates the user to the workout logging workflow.

BR-DASH-10: Display Recent Activity
The system shall display the user's most recent 3–5 logged workouts, including date and completion status.

BR-DASH-11: Quick Access to Activity Details
The system shall allow users to select a recent activity entry to view or edit its details.

BR-DASH-12: Display Progress Summary Cards
The system shall display summary cards showing weekly, monthly, and/or streak-based progress metrics.

BR-DASH-13: Display Notifications
The system shall display system-generated notifications relevant to the user's activity, program status, or data entry needs.

BR-DASH-14: Responsive Layout
The system shall render the dashboard in a responsive layout optimized for mobile, tablet, and desktop devices.

BR-DASH-15: Data Refresh
The system shall refresh dashboard data upon page load and after any user action that modifies underlying data.

BR-DASH-16: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon.

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BR-DASH-17: Navigation Menu Access
The system shall allow users to open the navigation menu from the dashboard via the menu icon.

BR-DASH-18: Notification Center Access
The system shall allow users to access the notification center by selecting the notification icon.

BR-DASH-19: Handle Missing Program Data
The system shall display a default message when no exercise program is selected.

BR-DASH-20: Handle Missing Weight Data
The system shall display a default message when no weight entries exist.

BR-DASH-21: Handle Missing Workout History
The system shall display a default message when no recent activity is available.

BR-DASH-22: Error Handling for Data Retrieval Failures
The system shall display an error message if dashboard data cannot be retrieved due to system or database errors.

BR-DASH-23: Calculate Weekly Progress
The system shall calculate weekly progress based on weight or workout metrics.

BR-DASH-24: Calculate Monthly Progress
The system shall calculate monthly progress based on weight or workout metrics.

BR-DASH-25: Calculate Consistency Metric
The system shall calculate the user's consistency metric based on completed vs. scheduled workouts.

BR-DASH-26: Display Workout Completion Status
The system shall display whether each recent activity entry was completed, skipped, or marked as a rest day.

BR-DASH-27: Restrict Dashboard Access
The system shall restrict access to the dashboard to authenticated users only.

BR-DASH-28: Session Validation
The system shall validate the user's session token before displaying dashboard data.

Menu

BR-MENU-01: Display Hamburger Menu Icon
The system shall display a hamburger menu icon in the top navigation bar on all authenticated pages.

BR-MENU-02: Open Menu Panel
The system shall open a navigation panel when the hamburger menu icon is selected.

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BR-MENU-03: Close Menu Panel
The system shall close the navigation panel when the user selects the close icon, taps outside the panel, or navigates to a menu item.

BR-MENU-04: Display Primary Navigation Items
The system shall display the following primary navigation options within the menu:

- Dashboard
- Exercise Programs
- Log Workout
- Weight Tracking
- Reports
- Profile
- Notifications
- Help
- Logout

BR-MENU-05: Navigate to Selected Page
The system shall navigate the user to the selected page when the user selects a menu item.

BR-MENU-06: Highlight Active Page
The system shall visually indicate the currently active page within the menu.

BR-MENU-07: Accessibility Compliance
The system shall allow the hamburger menu to be opened and navigated using keyboard and screen-reader accessibility standards.

BR-MENU-08: Responsive Behavior
The system shall adjust the menu layout based on device size, using a slide-out panel on mobile and a collapsible panel on larger screens.

BR-MENU-9: Session Awareness
The system shall restrict menu items requiring authentication unless a valid session is active.

Exercise Program

BR-EP-01: Exercise Program Page
The system shall provide users with access to a centralized view of all available exercise programs to support informed selection of a fitness plan.

BR-EP-02: Exercise Program Detail List
The system shall allow users to view detailed descriptions of each exercise program, including goals, duration, and included exercises, to support alignment with personal fitness objectives.

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BR-EP-03: Select an Exercise Program
The system shall allow users to select and activate a preferred exercise program to guide daily workout execution.

BR-EP-04: Update Exercise Program
The system shall allow users to update or change their selected exercise program at any time to accommodate evolving fitness needs.

BR-EP-05: Show Selected Program Progress
The system shall provide visibility into program progress (e.g., days completed, days remaining) to support user motivation and adherence.

Workout Program

BR-PROG-01: Display Program Detail Page
The system shall display a Program Detail page containing the program name, description, level, duration, and daily workout preview.

BR-PROG-02: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon.

BR-PROG-03: Display Program Name
The system shall display the name of the selected workout program.

BR-PROG-04: Display Program Description
The system shall display a brief description of the selected workout program.

BR-PROG-05: Display Program Level
The system shall display the difficulty level of the program.

BR-PROG-06: Display Program Duration
The system shall display the total duration of the program in weeks.

BR-PROG-07: Display Current Week and Day
The system shall display the user's current week and day within the selected program.

BR-PROG-08: Display Workout Structure Summary
The system shall display the workout structure for the selected day, including:

- Number of rounds
- Work interval duration
- Rest interval duration

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BR-PROG-09: Display Exercise List
The system shall display the list of exercises included in the workout for the selected day.

BR-PROG-10: Display Exercise Timing
The system shall display the timing for each exercise (e.g., "40 sec on / 20 sec off").

BR-PROG-11: Display Exercise Notes
The system shall display any exercise-specific notes (e.g., "incline or regular").

BR-PROG-12: Navigate Between Program Days
The system shall allow users to navigate to previous or next days within the program.

BR-PROG-13: Navigate Between Program Weeks
The system shall allow users to navigate to previous or next weeks within the program.

BR-PROG-14: Return to Program List
The system shall allow users to return to the list of available workout programs.

BR-PROG-15: Start Workout Action
The system shall provide a mechanism for users to begin the workout logging workflow from the Program Detail page.

BR-PROG-16: Validate Day Availability
The system shall prevent users from starting a workout for a day that is locked, unavailable, or future-dated.

BR-PROG-17: Handle Missing Program Metadata
The system shall display a default message if program metadata cannot be retrieved.

BR-PROG-18: Handle Missing Workout Day Data
The system shall display a default message if the selected day's work out details are unavailable.

BR-PROG-19: Error Handling for Data Retrieval Failures
The system shall display an error message if program data cannot be retrieved due to system or database errors.

BR-PROG-20: Restrict Program Access
The system shall restrict access to program details to authenticated users only.

BR-PROG-21: Validate Program Enrollment
The system shall validate that the user is enrolled in or has access rights to the selected program.

BR-PROG-22: Provide Accessible Labels
The system shall provide accessible labels for all program and exercise information to support screen readers.

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BR-PROG-23: Keyboard Navigation Support
The system shall allow users to navigate the Program Detail page using keyboard input only.

Log Workout

BR-LW-01: Log Workout Activity
The system shall provide users with the ability to log daily workout activity to support accurate tracking of exercise performance.

BR-LW-02: Log Reps, Weight, Duration
The system shall allow users to record exercise-specific metrics (e.g., sets, reps, weight, duration) to support detailed performance monitoring.

BR-LW-03: Current Day Workout
The system shall allow users to view the exercises associated with the current day of their selected program to support structured workout execution.

BR-LW-04: Edit Workout Log
The system shall allow users to modify or correct logged workout entries to ensure data accuracy.

BR-LW-05: Show Log Update Message
The system shall provide users with confirmation that a workout has been successfully logged to support user confidence and reduce duplicate entries.

BR-LW-06: Select Historical Workout by Date
The system shall allow users to view historical workout logs to support long-term performance analysis.

BR-LW-07: Display Workout Title
The system shall display the name of the workout (e.g., “Back and Shoulders”).

BR-LW-08: Display Workout Instructions
The system shall display instructional text associated with the workout, including rep ranges, tempo guidance, and rest period rules.

BR-LW-09: Display Exercise List
The system shall display all exercises included in the workout in the order defined by the program.

BR-LW-10: Display Sets and Reps
The system shall display the prescribed sets and rep ranges for each exercise.

BR-LW-11: Display Superset Relationships
The system shall indicate when exercises are part of a superset and show their pairing.

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BR-LW-12: Display Exercise Notes
The system shall display exercise-specific notes (e.g., "squeeze at the top for 2 seconds").

BR-LW-13: Display Logging Table
The system shall display a table for each exercise allowing users to enter reps and weight for each set.

BR-LW-14: Support Multiple Sets per Exercise
The system shall allow users to log metrics for each prescribed set individually.

BR-LW-15: Support Optional Weight Entry
The system shall allow users to leave weight blank for bodyweight exercises.

BR-LW-16: Navigate Between Exercises
The system shall allow users to scroll through or navigate between exercises within the workout.

BR-LW-17: Return to Program Day View
The system shall allow users to return to the Program Day Preview page.

BR-LW-18: Validate Numeric Inputs
The system shall validate that reps and weight entries contain valid numeric values.

BR-LW-19: Handle Missing Required Log Data
The system shall display an error message if required log fields are missing when the user attempts to save.

BR-LW-20: Handle Data Save Errors
The system shall display an error message if workout log data cannot be saved due to system or database errors.

BR-LW-21: Restrict Workout Access
The system shall restrict access to workout execution pages to authenticated users only.

BR-LW-22: Validate Program Enrollment
The system shall validate that the user has access to the selected workout program before displaying workout details.

BR-LW-23: Provide Accessible Labels
The system shall provide accessible labels for all exercise and logging fields.

BR-LW-24: Keyboard Navigation Support
The system shall allow users to navigate the workout logging interface using keyboard input only.

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BR-HELP-01: Display Help Page
The system shall display a Help page containing support contact information and frequently asked questions.

BR-HELP-02: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon

BR-HELP-03: Display Support Email
The system shall display the official support email address for users to contact NeuByte support.

BR-HELP-04: Support Email Link Behavior
The system shall allow users to select the support email address to open their device's default email client with a pre-addressed message.

BR-HELP-05: Display FAQ Section
The system shall display a Frequently Asked Questions section on the Help page.

BR-HELP-06: Display FAQ Questions and Answers
The system shall display each FAQ question along with its corresponding answer.

BR-HELP-07: Expand/Collapse FAQ Answers
The system shall allow users to expand or collapse FAQ answers to improve readability.

BR-HELP-08: Support Multiple FAQ Entries
The system shall support displaying multiple FAQ entries as defined by the business.

BR-HELP-09: Manage FAQ Content
The system shall allow authorized administrators to add, update, or remove FAQ questions and answers.

BR-HELP-10: Manage Support Contact Information
The system shall allow authorized administrators to update the support email address.

BR-HELP-11: Handle Missing FAQ Data
The system shall display a default message if no FAQ entries are available.

BR-HELP-12: Handle Data Retrieval Errors
The system shall display an error message if Help page content cannot be retrieved due to system or database errors.

BR-HELP-13: Restrict Access to Authenticated Users
The system shall restrict access to the Help page to authenticated users only.

BR-HELP-14: Provide Accessible Labels
The system shall provide accessible labels for all Help page content to support screen readers.

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BR-HELP-15: Keyboard Navigation Support
The system shall allow users to navigate the Help page using keyboard input only.

Weight Tracker

BR-WT-01: Body Weight Page
The system shall allow users to record their body weight to support long-term progress tracking.

BR-WT-02: Add/ Edit Weight by date
The system shall allow users to update or correct weight entries to ensure data accuracy.

BR-WT-03: Display Historical Weith Entries by Day
The system shall display historical weight entries in a chronological list to support trend visibility.

BR-WT-04: Weight Trend Graph
The system shall provide a visual representation (e.g., line graph) of weight trends over time to support user motivation and goal assessment.

BR-WT-05: Set Weight Goal
The system shall allow users to set weight-related goals to support personalized fitness planning.

Reports

BR-RPT-01: Display Report List Page
The system shall display a Report List page containing a list of available reports.

BR-RPT-02: Display Application Header
The system shall display the application header containing the NeuByte logo, notification icon, and menu icon

BR-RPT-03: Display Available Reports
The system shall display a list of all reports available to the user.

BR-RPT-04: Display Report Names
The system shall display the name of each report in the list.

BR-RPT-05: Support Multiple Report Entries
The system shall support displaying any number of report items as defined by the business.

BR-RPT-06: Select Report
The system shall allow users to select a report from the list.

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BR-RPT-07: Navigate to Report Detail Page
The system shall navigate the user to the corresponding report detail page upon selecting a report.

BR-RPT-08: Handle Missing Report List
The system shall display a default message if no reports are available.

BR-RPT-09: Handle Report Retrieval Errors
The system shall display an error message if the report list cannot be retrieved due to system or database errors.

BR-RPT-10: Restrict Report Access
The system shall restrict access to the Report List page to authenticated users only.

BR-RPT-11: User-Specific Report Filtering
The system shall ensure that users can only view reports associated with their own account.

BR-RPT-12: Provide Accessible Labels
The system shall provide accessible labels for all report items to support screen readers.

BR-RPT-13: Keyboard Navigation Support
The system shall allow users to navigate the Report List page using keyboard input only.

Notifications

BR-NOTE-01: Ongoing Notifications
The system shall provide users with notifications related to their fitness activity, program status, and account events.

BR-NOTE-02 Notification Icon Plus Count
The system shall display an unread notification count on the notification icon on all authenticated pages.

BR-NOTE-03: Notification List
The system shall allow users to view a list of all notifications, including read and unread items.

BR-NOTE-04: Action(s) that mark Notification Read
The system shall allow users to mark notifications as read by clicking or expanding the notification.

BR-NOTE-05: Closing / Marking Notification Read to prevent reloading
The system shall ensure that read notifications do not reappear as unread on subsequent page loads.

BR-NOTE-06: Notification Data Table
The system shall store all notifications in a persistent data store for audit and reporting purposes.

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BR-NOTE-07: Notification Triggers
The system shall generate notifications based on user actions (e.g., logging workouts) and system events (e.g., inactivity reminders).

BR-NOTE-08: Display Notification Date
The system shall display the date associated with each notification.

BR-NOTE-09: Display Notification Message
The system shall display the full text of each notification.

BR-NOTE-10: Display Notification List in Tabular Format
The system shall display notifications in a structured list or table with columns for date and message.

BR-NOTE-11: Sort Notifications by Date
The system shall sort notifications in descending chronological order by default.

BR-NOTE-12: Support Pagination or Scrolling
The system shall allow users to view additional notifications through scrolling or pagination when the list exceeds the visible area.

BR-NOTE-13: Handle No Notifications Available
The system shall display a default message when no notifications exist for the user.

BR-NOTE-14: Handle Notification Retrieval Errors
The system shall display an error message if notifications cannot be retrieved due to system or database errors.

BR-NOTE-15: Restrict Notification Access
The system shall restrict access to the notification center to authenticated users only.

BR-NOTE-16: User-Specific Notification Filtering
The system shall ensure that users can only view notifications associated with their own account.

User Profile

BR-PROF-01: Display User Profile Page
The system shall display a User Profile page that includes the user's email address and account-related settings.

BR-PROF-02: Edit Email Address (Optional Future Scope)
The system may allow users to update their email address in a future release.
[Marked as future scope to avoid scope creep.]

BR-PROF-03: Access Password Reset Functionality
The system shall provide a "Change Password" option within the User Profile page.

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BR-PROF-04: Validate Current Password
The system shall require users to enter their current password before allowing a password change.

BR-PROF-05: Validate New Password Complexity
The system shall validate that the new password meets defined complexity requirements.

BR-PROF-06: Confirm Matching New Passwords
The system shall validate that the new password and confirmation password match.

BR-PROF-07: Secure Password Update
The system shall update the user's password using a modern, salted, one way hashing algorithm consistent with NFR-SEC-02.

BR-PROF-08: Password Reset Confirmation
The system shall display a confirmation message when the password has been successfully updated.

BR-PROF-09: Error Messaging for Password Reset
The system shall display clear error messages for incorrect current password, weak new password, or mismatched confirmation fields.

BR-PROF-10: Transport Security
The system shall transmit all password reset data over HTTPS.

BR-PROF-11: Session Continuity
The system shall maintain the user's active session after a successful password change.

Non-Functional Requirements

NFR-PERF-01: Page Load Responsiveness
The system shall load all primary pages (Dashboard, Programs, Logging, Reports) within 2 seconds under normal usage conditions.

NFR-PERF-02: Data Submission Responsiveness
The system shall process user-submitted data (workout logs, weight entries, program updates) within 1 second of submission.

NFR-PERF-03: Lightweight Resource Usage
The system shall operate efficiently on typical consumer devices, including mobile phones, without requiring high CPU or memory usage.

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NFR-PERF-04: Concurrent User Handling

The system shall support up to ten concurrent users. without degradation, appropriate for a portfolio demonstration environment.

NFR-PERF-05: Cloud Resource Efficiency

The system shall utilize cloud resources in a cost-efficient manner, avoiding unnecessary compute, or storage consumption.

NFR-SEC-01: Encrypted Transport (SSL/TLS)

The system shall enforce HTTPS for all pages and API endpoints using a valid SSL/TLS certificate.

NFR-SEC-02: Secure Password Storage

The system shall store all user passwords using a modern, salted, one-way hashing algorithm (bcrypt).

NFR-SEC-03: Secure Session Management

The system shall store session tokens in HTTP-only, secure cookies to prevent client-side access.

NFR-SEC-04: Session Timeout

The system shall automatically expire inactive sessions after a defined period to reduce unauthorized access risk.

NFR-SEC-05: Input Validation

The system shall validate and sanitize all user inputs to prevent injection attacks (SQL injection, XSS, etc.).

NFR-SEC-06: Authentication Failure Handling

The system shall limit repeated failed login attempts to mitigate brute-force attacks.

NFR-SEC-07:

The system shall prevent automated login attempts through CAPTCHA enforcement.

NFR-SEC-07: Access Control

The system shall restrict access to authenticated pages unless a valid session token is present.

NFR-SEC-08: Error Message Safety

The system shall avoid exposing sensitive system details in error messages.

NFR-SEC-09: Data Privacy

The system shall ensure that user data is only accessible to the authenticated user who created it.

NFR-SEC-10: Logging of Security-Relevant Events

The system shall log authentication attempts, password changes, and account lockouts for audit and troubleshooting purposes.

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NFR-SEC-11: Secure Configuration
The system shall disable unnecessary services, ports, and default credentials in the hosting environment.

NFR-SEC-12: Dependency Security
The system shall use up-to-date, supported versions of all libraries and framework.

Business Requirements
Users seeking to improve their personal fitness often lack a centralized, easy-to-use tool for tracking exercise programs, logging workouts, monitoring progress, and visualizing long-term trends. NeuByte identified an opportunity to demonstrate how a structured, user-centered application can support these needs while highlighting modern system analysis, cloud architecture, and analytics capabilities.

The Fitness Tracking Application addresses this need by providing a streamlined, data-driven platform that enables users to manage their fitness journey effectively.

Business Objectives

Project-Level Objectives

- Demonstrate NeuByte's end-to-end execution of the Software Development Life Cycle through the delivery of a functional fitness-tracking application.
- Showcase NeuByte's standards for documentation, analysis, design, development, testing, and ongoing support.
- Provide a user-centered solution that supports personal fitness goals through intuitive workflows and accessible data insights.

Functional Objectives

- Enable users to create, update, and manage personalized exercise programs.
- Allow users to log exercise metrics associated with their selected programs.
- Provide users with the ability to track body weight and other personal metrics over time.
- Deliver periodic progress reporting (weekly, monthly, annually) to support long-term goal tracking and data-driven decision-making.

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Scope

In Scope

- User authentication and account management
- Exercise program creation, editing, and deletion.
- Workout logging and activity history
- Weight tracking and trend visualization
- Dashboard with personalized metrics and progress summaries
- Reporting (weekly, monthly, annual)
- Notifications and system alerts
- Responsive UI for mobile and desktop
- Hamburger menu navigation

Out of Scope

- Integration with third-party fitness devices or wearables
- Social features (leaderboards, friend lists, shared programs)
- Native mobile applications (iOS/Android)
- Advanced analytics or AI-driven recommendations
- Nutrition tracking
- Multi-tenant administration or enterprise RBAC

Success Metrics

The project will be considered successful when:

- Users can create and manage exercise programs without errors.
- Users can log workouts and weight entries consistently and accurately.
- Dashboard displays accurate, real-time summaries of user progress.
- Reports generate correctly for weekly, monthly, and annual periods.
- The system maintains secure authentication and session handling.
- The application loads primary pages within 2 seconds under normal conditions.
- The UI is fully responsive on mobile and desktop devices.

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7. Stakeholders

Primary Stakeholders

- End Users: Individuals tracking personal fitness goals.
- NeuByte Portfolio Reviewers: Hiring managers evaluating system analysis, documentation quality, and technical execution.

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Secondary Stakeholders

- NeuByte Development Team: Responsible for design, implementation, testing, and documentation.
- NeuByte QA Reviewers: Validate functional and non-functional requirements.

8. Assumptions

- Users have access to a modern web browser with JavaScript enabled.
- Users have stable internet connection.
- The application will be deployed in a cloud environment (e.g., Azure).
- Only ten concurrent users will access the system.

9. Constraint

- Limited scope appropriate for a portfolio demonstration project.
- No integration with external APIs or fitness devices.
- Cloud resources must remain cost-efficient.
- Security must follow modern standards (HTTPS, hashed passwords, secure sessions).

10. Success Criteria

The project will be considered successful when:

- Users can create and manage exercise programs without errors.
- Users can log workouts and weight entries consistently and accurately.
- Dashboard displays accurate, real-time summaries of user progress.
- Reports generate correctly for weekly, monthly, and annual periods.
- The system maintains secure authentication and session handling.
- The application loads primary pages within 2 seconds under normal conditions.
- The UI is fully responsive on mobile and desktop devices.

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11. Risks and Mitigation

Risk 1 — Data Loss or Corruption

Description: User-entered workout logs, weight entries, or program data may be lost due to system errors, deployment issues, or database misconfiguration.

Impact: Medium

Likelihood: Low

Mitigation:

- Implement automated database backups.
- Use transactional operations for data writes.
- Validate data before saving to prevent malformed entries.
- Evaluate database schema changes in a non-production environment.

Risk 2 — Unauthorized Access / Security Breach

Description: Weak authentication or insecure data handling could expose user accounts or personal fitness data.

Impact: High

Likelihood: Low

Mitigation:

- Enforce HTTPS for all traffic.
- Store passwords using salted, one-way hashing.
- Use secure, HTTP-only session cookies.
- Implement account lockout after repeated failed login attempts.
- Conduct basic security testing (input validation, session handling).

Risk 3 — Poor User Adoption Due to Usability Issues

Description: Users may find the interface confusing or unintuitive, reducing engagement and perceived value.

Impact: Medium

Likelihood: Medium

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Mitigation:

- Use a clean, mobile-first design with a hamburger menu.
- Provide clear navigation and consistent UI patterns.
- Conduct informal usability testing with sample users.
- Improve layout based on feedback.

Risk 4 — Performance Degradation

Description: Slow page loads or sluggish interactions may reduce user satisfaction.

Impact: Low

Likelihood: Medium

Mitigation:

- Optimize database queries and API calls.
- Use lightweight UI components.
- Cache frequently accessed data where appropriate.
- Set performance expectations (e.g., 2 second page loads).

Risk 5 — Feature Creep / Scope Expansion

Description: Adding too many features (nutrition tracking, social features, AI recommendations) may delay delivery or dilute the project's purpose.

Impact: Medium

Likelihood: High

Mitigation:

- Maintain a clear list for In Scope or Out-of-Scope.
- Prioritize features aligned with core business objectives.
- Defer enhancements to a future roadmap section.

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Risk 6 — Incomplete or Inaccurate Data Entry

Description: Users may forget to log workouts or enter inconsistent data, reducing the accuracy of reports.

Impact: Low

Likelihood: Medium

Mitigation:

- Provide notifications or reminders for missed logs.
- Validate data formats and required fields.
- Allow users to edit or correct past entries.

Risk 7 — Cloud Hosting or Service Outage

Description: Azure service interruptions may temporarily prevent access to the application.

Impact: Medium

Likelihood: Low

Mitigation:

- Use Azure services with built-in redundancy.
- Monitor service health.
- Provide user-friendly error messages during outages.

Risk 8 — Limited Testing Coverage

Description: As a solo portfolio project, testing may be less extensive than in a full development team.

Impact: Medium

Likelihood: Medium

Mitigation:

- Create a structured test plan and test cases.
- Perform functional, usability, and regression testing.
- Use automated tests where feasible.

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12. Revision History

Version	Date	Description	Author
1.0	04/11/2026	Initial draft	Gordon Neuls
1.1	04/18/2026	Add BR for Captcha and MFA	Gordon Neuls

13. Approval

Date	Role	Author
04/11/2026	Business System Analyst	Gordon Neuls
04/11/2026	Principle Software Engineer	Gordon Neuls

14. Glossary of Terms

Term	Definition
Account Lockout	A security control that temporarily disables a user account after repeated failed login attempts. Referenced in: "The system shall track failed login attempts and temporarily lock the account..." (BR-Login-14).
Activity Log	A record of completed workouts, including sets, reps, duration, and performance notes. Used for reporting and historical review.
Authenticated Page	Any page that requires a valid session token to access. Referenced in: "The system shall restrict access to authenticated pages..." (BR-Login-12).
Body Weight Entry	A user-submitted weight measurement stored chronologically for trend analysis. Referenced in: BR-WT-01 through BR-WT-05.
Business Requirements Document (BRD)	A formal document defining the objectives, scope, functional requirements, non-functional requirements, and constraints of the NeuByte Fitness Tracking Application.
Cloud Environment	The hosting platform (e.g., Azure) used to deploy the application and store user data. Referenced in: "The application will be deployed in a cloud environment (e.g., Azure)."
Concurrent Users	The number of users accessing the system simultaneously. Referenced in: NFR-PERF-04.
Cron Job / Scheduled Job	A server-side automated process that triggers system events such as inactivity reminders or periodic notifications
Dashboard	The primary landing page after login, displaying user metrics, program progress, recent activity, and notifications. Referenced in: BR-DASH-01 through BR-DASH-15.
Data Store	The persistent database, used to store user accounts, workouts, weight entries, programs, and notifications. Referenced in: BR-N-06.
Exercise Program	A structured set of workouts defined by goals, duration, and included exercises. Referenced in: BR-EP-01 through BR-EP-05.
Exercise Program Progress	The user's current position within a selected program (e.g., week/day). Referenced in: BR-DASH-05.
Failed Login Attempt	An unsuccessful authentication attempt(s) due to invalid credentials. Referenced in: BR-Login-14.
Fitness Report	A summary of user activity and trends over weekly, monthly, or annual periods. Referenced in: BR-RP-01 through BR-RP-06.
Hamburger Menu	A collapsible navigation menu accessible from all authenticated pages. Referenced in: BR-MENU-01 through BR-MENU-10.
Historical Data	Past workout logs, weight entries, or performance metrics used for reporting and trend analysis.

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Term	Definition
Last Login Timestamp	The date and time of the user's most recent successful login. Referenced in: BR-DASH-03.
Log Workout	The process of recording exercise metrics for a given day's workout. Referenced in: BR-LW-01 through BR-LW-06.
Navigation Panel	The slide-out or collapsible menu containing primary and secondary navigation items. Referenced in: BR-MENU-02.
Notification	A system-generated message related to user activity, program status, or account events. Referenced in: BR-N-01 through BR-N-07.
Notification Badge	A numeric indicator showing the count of unread notifications. Referenced in: BR-N-02.
Notification List	A page or panel displaying all read and unread notifications. Referenced in: BR-N-03.
Program Metadata	Descriptive information about an exercise program, including goals, duration, and included exercises. Referenced in: Scope → Exercise Program Management.
Progress Summary Cards	Dashboard UI components that represent weekly, monthly, or streak-based metrics. Referenced in: BR-DASH-12.
Read Flag (Notification)	A stored boolean value indicating whether the user has viewed a notification. Referenced in: BR-N-04 and BR-N-05.
Reports	Visual and tabular summaries of user performance, trends, and historical data. Referenced in: BR-RP-01 through BR-RP-06.
Session Token	A secure identifier stored in an HTTP-only cookie to maintain user authentication. Referenced in: BR-Login-08 and BR-Login-09.
Streak-Based Metrics	Performance indicators showing consecutive days or weeks of completed workouts. Referenced in: BR-DASH-12.
Trend Visualization	Graphical representation of weight or performance changes over time. Referenced in: BR-WT-04.
Unread Notification	A notification that has not yet marked as read and contributes to the unread count. Referenced in: BR-N-02.
User Profile	Basic account information such as email and weight history.
Weight Trend	The calculated direction of weight change (increase, decrease, no change). Referenced in: BR-DASH-07.
Workout Summary	A dashboard preview of the exercises scheduled for the current day. Referenced in: BR-DASH-08.

15. Appendix B: Document Cross Reference

See Traceability Matrix Excel Spreadsheet ..\RTM\NeuByteFitnessAppRequirementTracabilityMatrix.xlsx

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